

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q23-2360

Palm Village 138 kV
February 17, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2360. The Interconnection Customer proposes a 325.4 MW solar photovoltaic facility interconnecting at the Palm Village 138 kV substation in the Pennsylvania area. The request was received on November 26, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 497 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,318,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Palm Village – Johnston 138kV	Palm Village – Ganade 138kV line	146.6	183	100.0
Ganade – Palm Village 138kV	Palm Village – Johnston 138kV L	128.4	209	100.0
Paia Hydroelectric Plant – Jewell Branch 138kV	T Bone 7011 – Petrie 138kV	121.1	235	20.5
Paia Hydroelectric Plant – Reiner Rd 138kV	T Bone 7011 – Ptrie 138kV L	114.0	218	20.5
Coconut Groe – Johnston 138kV	Ganade – Palm Village 138kV	108.3	224	100.0
Reiner Road – Medway 138kV	T Bone 7011 – Petrie 138kV line	106.8	192	20.5
Medway – Blue Goose 138kV L	Marcus Hook Energy Lp – Tombstone 500kV	106.5	247	14.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Palm Village – Johnston 138kV	Rebuild line with higher-capacity conductor	\$2,198,000	100% (\$2,198,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ganade – Palm Village 138kV	Rebuild line with higher-capacity conductor	\$4,029,000	100% (\$4,029,000)
Paia Hydroelectric Plant – Jewell Branch 138kV	Rebuild line with higher-capacity conductor	\$2,956,000	63% (\$1,852,000)
Paia Hydroelectric Plant – Reiner Rd 138kV	Rebuild line with higher-capacity conductor	\$1,271,000	63% (\$797,000)
Coconut Grove – Johnston 138kV	Reconductor line	\$3,960,000	100% (\$3,960,000)
Medway – Reiner Rd 138kV	Reconductor line	\$575,000	63% (\$361,000)
Blue Goose – Medway 138kV	Reconductor line	\$243,000	50% (\$121,000)
	Total allocated to project		\$13,318,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.